TatvaAI: feature list in simple words

What the project does

Every year, colleges ask AICTE to renew their approval. They upload documents like fire safety and land certificates. Officers check them by hand, which is slow, and fake or expired papers can slip through.

TatvaAI reads those documents, checks them, and warns the officer about problems. It shows proof for every warning. It gives each college a risk colour: GREEN (no risk), AMBER (medium risk) or RED (high risk). It also makes a list of which colleges inspectors should visit first.

The software never approves or rejects anyone. A human officer decides, and must write a reason.

Status words: **Done** = works for real. **Fake for now** = the screen can be built, but the data is made up. **To build** = not started.

1. For the college

Feature	What it means	Status
Login	Each person signs in as college, officer or inspector	Done
Start an application	Pick the year and type (renewal, more seats, name change)	Done
Upload documents	PDFs, photos and the faculty list are accepted and safety-checked	Done
Recognise documents	The system says what each file is, like "fire certificate, 97% sure"	Done
Pre-check	A plain to-do list of what to fix before submitting	Done
Submit	Locks the application; the officer can reopen it once	Done

2. Reading the documents

Feature	What it means	Status
Read the fields	Finds each value (dates, numbers, names) and remembers where it sits on the page	Done
Check digital signatures	Sees if a document was changed after it was signed, and shows what changed	Done
Read QR codes	Compares the QR with the printed text	Done
Read photo details	Where and when a photo was taken	Done
Read the faculty list	Turns the spreadsheet into records	Done

3. Checking with the original source

Feature	What it means	Status
Fake government sources	Pretend e-stamp, DigiLocker, CA, fire, building and AICTE systems	Done
Ask the source	Ask the right source if each document is real	Done
Six answers	Verified, mismatch, expired, pending, source down, needs manual check	Done

Feature	What it means	Status
Source is down	The case goes to a person instead of getting stuck	Done
Reuse last year	A document already verified last year is not checked again	Done

4. The five checkers (they raise warnings, called flags)

Checker	What it looks for	Status
Source checker	Source says the document is fake, expired or unreachable	Done
Authenticity checker	Edited after signing, QR mismatch, copied documents	Done
Completeness checker	Missing, unsigned or expired documents and fields	Done
Consistency checker	The same fact disagrees between documents, years or the college website	Done
Rules checker	Not enough teachers, rooms, computers or building area; photos taken far from the campus	Done

Every warning has a plain message, the proof it is based on, and the Handbook clause where it applies.

5. Rules, risk and inspections

Feature	What it means	Status
Rules in a file	AICTE norms are kept in editable files, not hidden in code	Done
Evidence collection	All facts gathered with where each came from	Done
Risk score and colour	Warnings add up to GREEN, AMBER or RED	Done
Inspection list	RED colleges ranked, with reasons and an on-site checklist	Done
Rule editor for the officer	Test a rule change before turning it on	Done

6. Officer and inspector screens

Feature	What it means	Status
Review queue	All colleges, highest risk first	Done
Review page	Summary, warnings with proof, verification, rules, map	Done
Show the proof	Opens the document page with the exact spot highlighted	Done
Accept or dismiss	Officer decides each warning and must write a reason; there is no reject button	Done
Alerts	Missing info, expiring documents, suspicious findings	Done
PDF report	A full report with the proof images	Done

7. Safety and trust

Feature	What it means	Status
Each user sees only their data	Colleges see only their own application	Done
Tamper-proof history	Every action is logged in a chain that shows if anyone altered it	Done
Document viewing log	Who opened which document	Done
Extra login step	Code from a phone app, for officers	Done
Only invented data	All names, colleges and sources are fake and labelled	Done

What exists today

Everything above is built and covered by the automated tests (backend suite plus the frontend build and smoke test), working against three made-up colleges: A is clean, B has two problems, C has many hidden ones (a land record edited after signing, 12 duplicate teachers, a copied photo).

The inspector now has a per-document on-site checklist: for every flagged item they record **Confirmed** (the issue is real) or **Cleared** (false alarm) with a written note, and a progress bar shows how much of the visit is done. This is separate from the officer's earlier accept/dismiss screening.

Optional extras stay switched off by default (scanned-page OCR, PC counting in lab photos, a local model that drafts rules). All sources are fake and labelled MOCK.